

Comments on Proposal: Increase k to 1000

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Takeaways

- This report studies the period around the increase in k from 150 to 500, to inform the discussion about a possible increase from 500 to 1000.
- Since one motivation for increasing k is to increase decentralization by pushing delegation from large pools to smaller ones, our analysis follows a fixed cohort of 836 pools with positive stake in epoch 228 that remained present in the epoch-249 and epoch-263 exports.
- Consequently, this approach does not provide a comprehensive decentralization analysis of the entire pool landscape. Instead, it isolates whether pools that existed prior to the increase in k improved over time.
- Data shows that small and medium pools from epoch-228 cohort gained delegators and stake between epochs 228 and 249/263. This is consistent with an improved position for some previously low-stake pools.
- The improvement was not universal: many very small pools, especially those below 1.5M ADA in stake, left the cohort after epoch 228.
- These gains should be interpreted carefully. Total active stake, the number of delegators, and ADA price also increased over the same period, so part of the observed improvement may reflect broader ecosystem growth rather than only the change in k .
- To further assess this, we compare network-level growth with growth inside our fixed cohort. Network-level growth reflects the net increase in total system delegators and stake, while cohort growth may come from new delegators, migration from other pools, or both.
- The comparison shows that small and medium pools grew faster than the network in both delegators and stake. Larger pools grew more slowly in delegators, and the largest stake buckets lost stake.
- This pattern is consistent with some reallocation of stake from larger pools toward smaller pools. For delegator counts, however, the evidence of large-to-small migration is weaker: most buckets gained delegators, so much of the growth likely reflects new inflows or inflows from outside the fixed cohort.
- Overall, increasing k appears to have helped small and medium pools, especially in terms of stake. However, its positive effect was likely amplified by favorable market conditions.

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1 Context

In this brief report, we analyze the potential effects of increasing the parameter k from 500 to 1000. To inform this analysis, we use data from the previous increase in k , from 150 to 500, which occurred at the start of epoch 234, to better understand how that change affected delegation centralization.

The most direct expected consequences of an increase in k are the following:

- The saturation point decreases.
- As a result, larger pools close to saturation lose rewards per unit of stake.
- Delegators seeking to optimize their rewards have an incentive to migrate toward less delegated pools.
- This migration should benefit smaller pools and, in principle, increase decentralization.
- However, large operators may respond by creating additional pools and reallocating pledge and delegation across them, thereby reducing their losses. In this case, we may observe a larger number of highly delegated pools but without a corresponding increase in true decentralization, since control may remain concentrated among a few operators.
- This strategy is especially relevant for operators who control a large share of their own delegation or have the capacity to influence the delegation decisions of their delegators.

These considerations raise the question of whether an increase in k actually leads to greater decentralization. Previous analyses make a comparison of data before the change in k and after it (e.g., <https://www.iog.io/news/not-long-till-d-0-day> considers epochs 228 and 249). By considering the full set of pools in each epoch, they provide a broad view of the pool landscape, how delegation is distributed across pools, and measure decentralization in each epoch.

However, decentralization measures such as the Nakamoto coefficient have an important limitation: they should ideally account for multi-pool operators. Additionally, a simple comparison across epochs may also capture effects unrelated to the change in k . For example, the ADA price was approximately \$0.11 at epoch 228, \$0.93 at epoch 249, and \$1.35 at epoch 263 (source: <https://cexplorer.io/>). This substantial increase in the token price may help explain the growth in total staked ADA, from 17.2B ADA in epoch 228 to 22.3B ADA in epoch 249 and 22.7B ADA in epoch 263. Such growth could itself have encouraged the creation of new pools, independently of the incentives generated by the change in k .

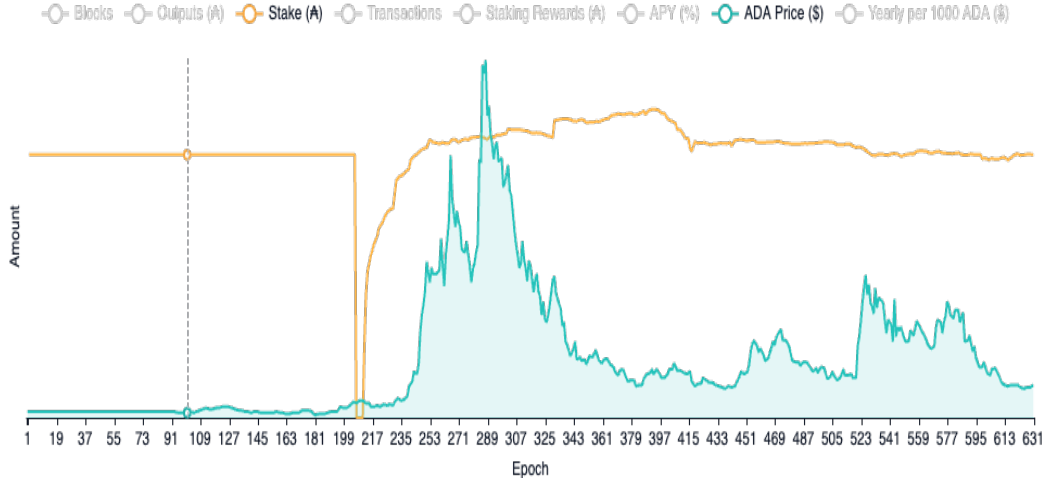


Figure 1: Historical stake and ADA price. Source: <https://cexplorer.io/>.

All in all, further evidence is needed to determine whether the previous increase in k delivered benefits to the pool landscape. This report aims to address this gap by introducing an alternative analytical approach.

2 This report

We take a different approach to complement previous analyses. Instead of comparing the full pool landscape across epochs, we follow a fixed cohort of pools that already existed before the previous increase in k , from 150 to 500. The goal is to study whether those pools improved over time in terms of stake and number of delegators.

This approach is motivated by one of the main arguments for increasing k : a lower saturation point should make large pools less attractive at the margin and create room for smaller pools to attract delegation. A natural question, therefore, is whether pools that were small/medium before the previous increase in k improved after the change.

To examine this, we focus on pools that appear in epoch-228, epoch-249, and epoch-263 exports and had positive stake in epoch 228, giving us a cohort of 836 surviving pools. Our main comparison is between epoch-228 versus epoch-249 but in some cases we also look at epoch 263 since it provides additional time for pools and delegators behavior to adjust.¹

By restricting our analysis to pools that existed before the change and survived through subsequent exports, we lose a comprehensive view of the changing pool landscape. However, this approach allows us to focus on our primary goal: determining whether pre-existing pools improved over time.

Notice that the evidence we present below is descriptive and does not fully identify the effect of the increase in k . Other factors may also have contributed to the observed increase in delegators or stake.

2.1 Summary of data

The following table summarizes the staking framework at each epoch.

¹At the same time, the longer the window, the greater the risk that observed changes reflect other factors besides the change in k , such as broader ecosystem growth or changes in the ADA price.

Epoch	Active pools	Total supply (B ADA)	Total stake (B ADA)	k	z_0 (M ADA)	Delegators	Avg. delegation (ADA)
228	1,086	32.03	17.20	150	213.53	65,987	248,472
249	1,583	32.19	22.30	500	64.38	208,520	104,079
263	1,617	32.52	22.70	500	65.04	416,288	48,001

Table 1: Epochs summary. Sources: Total stake from Cexplorer; rest of data from Koios export.

3 Results

Main Results:

The following Figure 4 shows how the distribution of pools by number of delegators evolved across the three epochs. In particular, it shows that pools in our cohort gained delegators. The number of pools with fewer than 100 delegators is the only bucket that declines, while the number of pools in all larger-delegator buckets increases, especially in the highest buckets. This is, there are now fewer small pools, meaning the average pool has a larger base of delegators. Nevertheless, the low-delegator bucket remains by far the largest group, indicating that most pools in the cohort continued to have a relatively small delegator base.

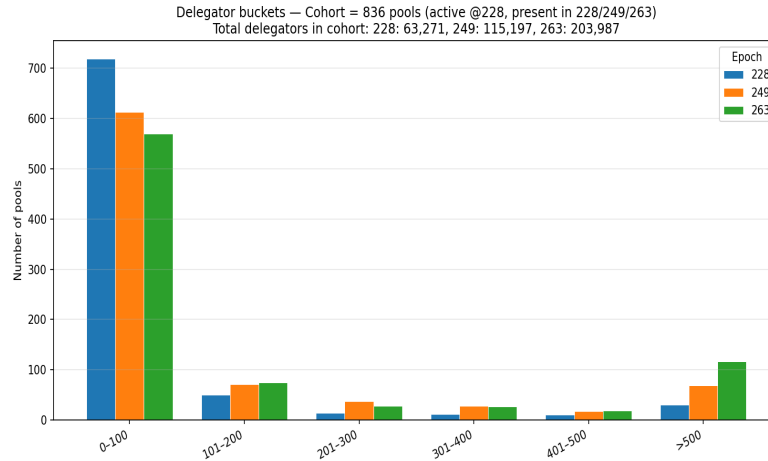


Figure 2: Distribution of pools by number of delegators across epochs. The comparison is restricted to pools that were active in epoch 228 and remain active in 249 and 263.

However, the figure do not allow us to distinguish whether these additional delegators were migrating from larger pools in response to the increase in k , or whether they were mainly new delegators entering the ecosystem. Since total active stake increased substantially over the same period (see Figure 1), it is reasonable to expect that a significant share of the additional delegation came from new stakers rather than only from reallocation across existing pools.

One way to further assess this pattern is to compare growth at the network level with growth within each stake bucket of our cohort. Network-level growth reflects the net increase in delegators and stake in the system, while growth within a cohort bucket may come from two sources: new delegators entering the network or existing delegators migrating from other pools. The following

¹The mismatch between the aggregate stake reported by Cexplorer and the sum of stake in our Koios exports indicates that the export is not a complete accounting of all stake in each epoch.

table compares these changes for the number of delegators. The comparison shows that small and medium pools in the cohort gained delegators faster than the network as a whole, while larger pools grew more slowly or, in the largest bucket, lost delegators. This pattern is consistent with smaller pools becoming relatively more attractive after the increase in k , but it provides only weak evidence of direct large-to-small delegator migration, since most buckets gained delegators and the only net loss comes from the largest bucket.

Bucket	Δ delegators	Bucket %	Network %	Above network?
0–1M	+11,451	+497%	+216%	Yes
1M–5M	+12,305	+357%	+216%	Yes
5M–15M	+8,620	+280%	+216%	Yes
15M–40M	+14,167	+201%	+216%	No
40M–100M	+10,179	+69%	+216%	No
>100M	−5,193	−15%	+216%	No

Table 2: Cohort delegator growth by stake bucket fixed at epoch 228 (period 228→249). Bucket cutoffs use each pool’s total stake at epoch 228 (0–1M, . . . , >100M ADA). A pool stays in its initial bucket for this table even if later stake moves it to another tier. Network delegator growth uses official totals ($\approx 66,000 \rightarrow 209,000$, +216%). Cohort: pools active at 228 present in epochs 228, 249, and 263.

We also want to know what happened with the distribution of stake and not only with the number of delegators. Figure 3 shows how the active-stake distribution changed within our cohort. The number of pools with less than 5M ADA decreases, while the number of pools in moderate and higher stake buckets increases.

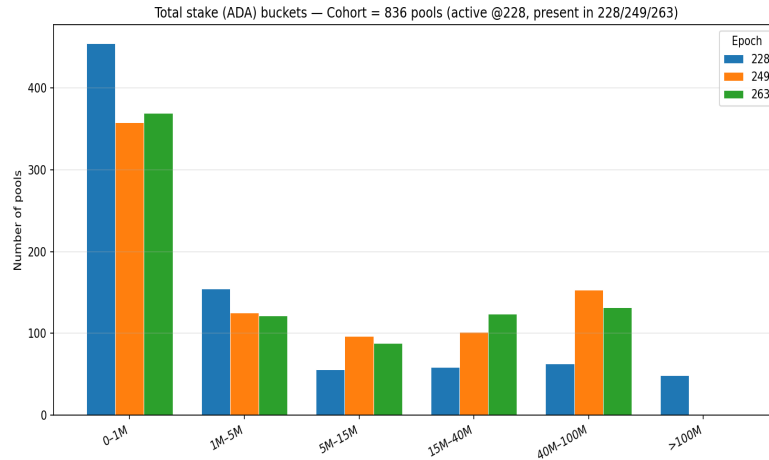


Figure 3: Distribution of pools by active stake across epochs. The sample is restricted to cohort of pools active in epoch 228 .

The next table makes the same comparison for stake levels as Table 2. We again observe that stake increased faster in the small and medium pools of our cohort than in the network as a whole. In contrast, the larger buckets, especially pools above 40M ADA in epoch 228, lost stake. This provides stronger evidence of possible stake reallocation than the delegator-count comparison, since the stake losses in large buckets are substantial. However, the cohort as a whole lost 2.18B ADA, so the gains of small and medium pools cannot be interpreted as a full redistribution within the cohort.

Part of the stake lost by large pools may have moved to pools outside the cohort, including newly created pools, possibly as part of operators’ adjustment to the lower saturation point.

Bucket	Δ stake (B ADA)	Bucket %	Network %	Above network?
0–1M	+1.19	+1359%	+30%	Yes
1M–5M	+1.04	+290%	+30%	Yes
5M–15M	+0.73	+137%	+30%	Yes
15M–40M	+0.71	+46%	+30%	Yes
40M–100M	−0.96	−24%	+30%	No
>100M	−4.89	−60%	+30%	No

Table 3: Cohort stake growth by stake bucket fixed at epoch 228 (period 228→249). Same initial buckets and cohort as Table 2. Network stake growth uses official totals (17.2 → 22.3 B ADA, +30%). Bucket Δ stake sums pool-level active stake within each fixed group.

A further caveat is that the observed pattern should not be read as a direct accounting of delegator movements from large incumbent pools to smaller incumbent pools. In the delegator-count comparison, most buckets gained delegators, and only the largest bucket shows a net loss. Therefore, the delegator gains of smaller pools cannot be mainly explained by net migration from the largest pools within our cohort. A large part of this growth likely reflects new delegators entering the system, or delegators coming from pools outside the fixed cohort.

The stake comparison provides stronger evidence of possible reallocation, since the larger stake buckets lose substantial stake while smaller and medium buckets gain stake. However, the cohort as a whole loses stake, so the gains of smaller pools cannot be interpreted as a full redistribution within the cohort. Part of the stake lost by large incumbent pools may have moved to pools outside the cohort, including newly created pools.

Takeaway: Delegator growth in the surviving cohort

Between epochs 228 and 249/263, surviving pools from the epoch-228 cohort gained delegators on net. However, this growth does not by itself identify whether delegators came from large pools, new entrants, or pools outside the cohort.

Takeaway: Improved position of some surviving pools

Small and medium pools in the surviving epoch-228 cohort increased their stake over the period studied. This is consistent with an improved economic position for some previously low-stake pools.

Takeaway: Other drivers of growth are relevant

During the period analyzed, the number of pools, the number of delegators, total active stake, and ADA price all increased. Therefore, the observed improvement of our cohort may also reflect broader ecosystem growth or other factors.

Takeaway: Evidence consistent with some stake reallocation

When pools are grouped by their epoch-228 stake, small and medium pools grew faster than the network in stake, while larger pools, especially those above 40M ADA, lost stake. This pattern is consistent with some reallocation of stake from larger pools toward smaller pools, although part of the lost stake may have moved outside the fixed cohort.

For delegator counts, the evidence is weaker. Most buckets gained delegators, and only the largest

bucket shows a net loss. Moreover, this loss is much smaller than the gains observed in the other buckets, suggesting that much of the delegator growth came from new inflows rather than from net migration out of the largest cohort pools.

Secondary Result:

The previous evidence shows that pools of our cohort gain delegators and stake, specially the small and medium ones. The next question is what type of delegation growth this represents.

The following figures focus on this point. Figure 4 shows that, over time, more pools in the cohort fall into buckets with low average stake per delegator, while fewer pools exhibit a high average stake per delegator. Figure 5 shows that the total number of delegators is increasingly concentrated in lower average-delegation buckets. These patterns are consistent with many additional delegators being relatively small delegators.

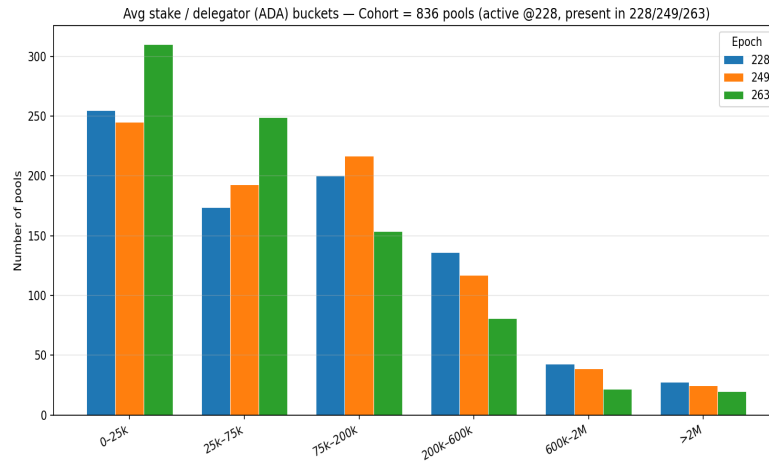


Figure 4: Distribution of pools by average staked per delegator across epochs.

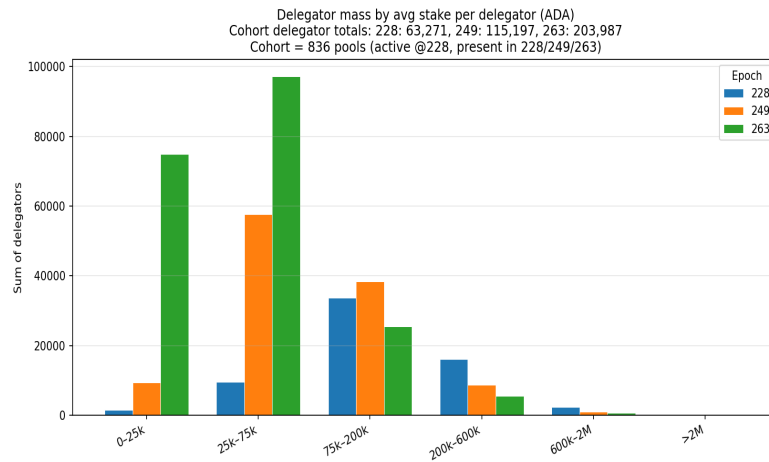


Figure 5: Total delegators in the 836-pool cohort (epochs 228, 249, 263), summed by average stake per delegator bucket.

Takeaway: Average stake per delegator decreases

The decline in average stake per delegator is consistent with many additional delegators being relatively small delegators.

3.1 Pools out of the relevant cohort

In the previous analysis, we focused on a cohort of 836 pools that were active in epoch 228 and remained present through epoch 263.

This leaves 250 pools from the original epoch-228 cohort ($= 1086 - 836$). Of these, 196 are already absent from the epoch-249 data, while the remaining 54 are present in epoch 249 but absent from the epoch-263 data. Why did these pools exit the cohort? Several explanations are possible. Some pools may no longer have been economically viable, while others may have been closed by operators who subsequently opened new pools. The following tables provide descriptive evidence on the characteristics of these 250 pools.

Category	Pools	% of 196	Stake @228	% group stake	Mean stake @228	Avg. Num. delegators
Tiny (< 1.5 M)	171	87.2%	0.03 B	1.1%	0.16 M	3.4
Small (1.5M–5M)	4	2.0%	0.01 B	0.3%	2.11 M	19.5
Medium (5M–25M)	2	1.0%	0.02 B	0.8%	9.64 M	58.5
Large (> 25 M)	19	9.7%	2.41 B	97.8%	126.7 M	86.8
Total	196	100.0%	2.46 B	100.0%	12.6 M	12.4

Table 4: Pools active at epoch 228 that left before epoch 249 ($n = 196$). Total stake @228: 2.46 B ADA (15.0% of all 1,086 pools @228).

Category	Pools	% of 54	Stake @228	% group stake	Mean stake @228	Avg. Num. delegators
Tiny (< 1.5 M)	49	90.7%	0.01 B	9.1%	0.17 M	3.8
Small (1.5M–5M)	3	5.6%	0.01 B	11.6%	3.49 M	15.7
Medium (5M–25M)	1	1.9%	0.01 B	6.5%	5.87 M	41.0
Large (> 25 M)	1	1.9%	0.07 B	72.7%	65.61 M	8.0
Total	54	100.0%	0.09 B	100.0%	1.67 M*	5.3*

Table 5: Pools in the epoch-249 export that left before epoch 263 ($n = 54$). Total stake @228: 0.09 B ADA (0.6% of all 1,086 pools @228)

These data indicate that most of the pools leaving the cohort were tiny pools, with less than 1.5M ADA in stake and only a small number of delegators, possibly just the operators behind the pools. This suggests that the improvement observed among surviving pools was not universal. Very small pools may have remained unable to compete even during the period following the increase in k , making it optimal for some operators to close them or reorganize their delegation.

The data also show that some large pools left the cohort. One possible explanation is that these pools were part of operators’ post-change optimization strategies, for example through pool closure, reallocation of pledge, or the creation of new pools. Importantly, however, the evidence does not suggest that small-to-medium pools systematically exited the staking ecosystem.

Takeaway: Very small pools were not rescued

Many pools with less than $1.5M$ ADA in stake left the cohort between epochs 228 and 249/263. This suggests that the improvement observed among surviving pools did not extend to many very small pools.